Mukul Rayana

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Technical Skills

Languages & Tools: Python, C#, SQL; data structures and algorithms; AI coding assistants
AI/ML: Generative AI, RAG, agentic systems, LangGraph, NLP, QLoRA, PyTorch, TensorFlow, scikit-learn
Backend & Deployment: REST APIs, FastAPI, Docker, GitHub Actions CI/CD, AWS (EC2, S3)
Data & Search: PostgreSQL, MongoDB, SQLite, pgvector, Pinecone, FAISS

Education

University of Maryland, College Park Aug 2024 - May 2026
M.S., Applied Machine Learning College Park, MD
Coursework: Deep Learning, Computer Vision, Computing Systems for ML, Cloud Computing, Optimization.

SRM Easwari Engineering College Dec 2020 - Apr 2024
B.Tech., Artificial Intelligence & Data Science Chennai, India
Certifications: Data Structures & Algorithms Specialization, UC San Diego; IBM Data Science Professional Certificate.

Experience

BeeBox Studios Pvt Ltd | IIT Madras Research Park Apr 2023 - Apr 2024
AI Research Intern Chennai, India

- Cut P95 inference latency 3x, from **185 ms to 62 ms**, with ONNX export and INT8 quantization, so a BERT question-answering model (**83.2% F1**) could run on limited GPU hardware for XR headsets.
- Delivered multi-peer WebRTC support a day ahead of a four-day client deadline after taking over an unfamiliar JavaScript and Python module, including ICE candidate exchange and socket signaling.
- Connected Python inference to the Unity/C# client over TCP as the only ML engineer, and deployed to AWS EC2 with Docker and GitHub Actions CI/CD, supporting **six production releases**.

Projects

Irminsul: Public Q&A Service with Cited Answers | Live Demo | GitHub Nov 2025 - Present

- Built and run a public service used by **15 Genshin Impact players**. It handled **210 requests/s at 0.45 s p95** with zero errors in a 50-user load test. Exact lookups skip the LLM and return in about 50 ms.
- Built hybrid search (BM25, embeddings, cross-encoder re-ranking on ONNX) that finds the right source in the top 8 for **94.6%** of 93 test questions. The live service passes **113 of 118** end-to-end tests, including all 19 safety probes.
- Automated a daily GitHub Actions pipeline that rebuilds the data and blocks publishing unless leak, search-quality and safety checks pass.

PharmaChain: Cold-Chain Shipment Incident Response | Live Demo | GitHub Apr 2026

- Won **1st place ($4,000)** at the UMD Smith Agentic AI Challenge with a team of five. The eight-agent system goes from shipment alert to approved action plan in **12 to 15 seconds**, versus the usual 6 to 8 hours.
- Built the Route and Triage agents and their REST API. Added failover across four LLM providers with a rule-based fallback, so the system keeps running through a full provider outage.

RECON: Stale Evidence Detection for Research Search | Live Demo | GitHub Oct 2025 - Jan 2026

- Built a four-agent LangGraph pipeline that flags evidence replaced by newer work, catching **44%** of stale claims on a 130-question benchmark versus **0%** for a single-pass RAG baseline, with a 2% false-positive rate.
- Fixed a verdict-ordering bug that hid contradiction checks, traced the remaining 0% catch rate to retrieval returning related rather than opposing papers, and archived a patch that inflated the score.

DocPilot: Secure Document Q&A API | GitHub Apr 2026

- Built a FastAPI service with role-based access, audit logging, PII redaction and prompt-injection checks. Cut model size **4x** and per-inference CPU latency **2 to 3x** with ONNX INT8 quantization.

Publication

Voice-Driven Panoramic Imagery: Real-Time Generative AI for Immersive Experiences | IEEE Xplore IDCIoT 2024

- Co-author. Built speech-to-360-degree image generation with Whisper and Stable Diffusion at **2.3x** throughput.